

1.4 Matrix Group

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Matrix Group

Notation : In $\mathbb{C}^n$, $\bar{e}_i = (0, \dots, 1^{th}, 0 \dots)$ is a vector space basis element.

* An endomorphism α of $\mathbb{C}^n$ is defined as

$$\alpha \bar{e}_i = \sum_{j=1}^n a_{ji} \bar{e}_j$$

where $\alpha = (a_{ij}) \in M_n(\mathbb{C})$.

(We can think of $\alpha \bar{e}_i$ as the i^{th} column vector.)

We use α for a matrix as well as for the endomorphism.

We define multiplication of two endomorphisms α, β with corr. matrices $(a_{ij}), (b_{ij})$ and construct (c_{ij}) where

$$c_{ij} = \sum_{k=1}^n a_{ik} b_{kj}$$

Lemma 0.1. $M_n(\mathbb{C}) \cong \mathbb{C}^{n^2}$ as vector spaces.

Proof. Construct the function

$$\begin{aligned} f : M_n(\mathbb{C}) &\rightarrow \mathbb{C}^{n^2} \\ (a_{ij}) &\mapsto (b_{ij}) \end{aligned}$$

where $b_{i+(j-1)n} = a_{ij}$.

Observe as j goes from 1 to n , $(j-1)n$ goes from 0 to $(n-1)n$ and as i goes from 1 to n as well, we get $i + (j-1)n$ go from 1 to n^2 .

Observe f is surjective by construction and if $a_{ij} \neq a_{i'j'}$ then $b_{i+(j-1)n} \neq b_{i'+(j'-1)n}$ hence f is bijective.

Observe f is linear hence $M_n(\mathbb{C}) \cong \mathbb{C}^{n^2}$. □

Remark 0.1. Observe we can equip $M_n(\mathbb{C})$ with a topology by ensuring the given map above to be a homeomorphism. By constructing

$$\tau_{M_n(\mathbb{C})} = \{f^{-1}(CU) \mid U \in \tau_{\mathbb{C}^n} \times \tau_{\mathbb{C}^n}\}$$

Lemma 0.2. Let T be any topological space,

$$\phi : T \rightarrow M_n(\mathbb{C})$$

ϕ is continuous if and only if $\pi_{ij} \circ \phi$ is continuous.

Proof. Only if condition is trivial.

If condition:

$$\pi_{ij} \circ \phi : T \rightarrow \mathbb{C}$$

is continuous.

Now $\prod_{1 \leq i, j \leq n} \pi_{ij} \circ \phi : T \rightarrow \mathbb{C}^{n^2}$ is continuous by product topology (or box topology as $\mathbb{C}^{n^2}$ is finite Cartesian product). $\square$

as $\pi_{ij}^{-1}(U)$ for U open in $\mathbb{C}$ forms a subbasis for product topology on $\mathbb{C}^{n^2}$.

$\square$

Corollary 0.1. Product of two matrices is continuous

Proof.

$$\text{Pf: } M_n(\mathbb{C}) \times M_n(\mathbb{C}) \xrightarrow{q} M_n(\mathbb{C})$$

$$\begin{array}{ccc} M_n(\mathbb{C}) \times M_n(\mathbb{C}) & \xrightarrow{f \times f} & \mathbb{C}^{n^2} \times \mathbb{C}^{n^2} \\ \downarrow q & & \downarrow \rho \\ M_n(\mathbb{C}) & \xrightarrow{f} & \mathbb{C}^{n^2} \end{array}$$

where $q : M_n(\mathbb{C}) \times M_n(\mathbb{C}) \rightarrow M_n(\mathbb{C})$,

$$(\sigma, \tau) \mapsto \sigma\tau$$

is matrix multiplication. And ρ is a polynomial to $p((b_{ij}), (b'_{ij})) = (c_{ij})$ where

$$c_{ij} = \sum_{k=1}^n b_{ik} b'_{kj}.$$

Hence ρ is continuous. Therefore for $q = \rho \circ (f \times f)$ is continuous. For any open set in $M_n(\mathbb{C})$, U , we express $U = f^{-1}(O)$ for some open set O in $\mathbb{C}^{n^2}$. $\square$

$$\mathbb{C}^{n^2}. \text{ Hence } q^{-1}(U) = q^{-1}(f^{-1}(O)) = (f \circ q)^{-1}(O)$$

which is open as $f \circ q$ is continuous.

$\square$

Notation: ${}^t A$ is transpose of a matrix and $\overline{A}$ is complex conjugate of a matrix.

Lemma 0.3. Transpose and complex conjugate are homeomorphism.

Proof. If $h : M_n(\mathbb{C}) \rightarrow M_n(\mathbb{C})$,

$$M_n(\mathbb{C}) @>h>> M_n(\mathbb{C}) @VVA \mapsto^t AV @VVA \mapsto \overline{AV}$$

we can construct $r : \mathbb{C}^{n^2} \rightarrow \mathbb{C}^{n^2}$ which is a rearrangement of indices such that

$$b_{i+j-n} = b_{j+i-c-1n} \quad \text{which is a homeomorphism.}$$

Hence, $h = f^{-1}rf$ is a homeomorphism.

If $k : M_n(\mathbb{C}) \rightarrow M_n(\mathbb{C})$ is homeomorphic

$$M_n(\mathbb{C}) @>k>> M_n(\mathbb{C}) @VVA \mapsto \overline{AV} @VVA \mapsto^t AV$$

□

as k_{ij} is homeomorphic for $1 \leq i, j \leq n$.

Property of matrices

$$\begin{aligned} (\alpha\beta)^t &= \beta^t \alpha^t \\ \frac{d(\alpha\beta)}{dt} &= \frac{d\alpha}{dt} \cdot \beta + \alpha \cdot \frac{d\beta}{dt} \end{aligned}$$

* A regular/non singular $n \times n$ matrix, if it has an inverse i.e. if there exists a matrix σ^{-1} such that $\sigma\sigma^{-1} = \sigma^{-1}\sigma = E$, where E is the unit matrix of degree n .

A necessary and sufficient condition for σ to be regular is $\det \sigma \neq 0$.

Remark 0.2. If σ is a regular matrix

1. iff σ^{-1} its reciprocal endomorphism exists
2. then ${}^t(\sigma^{-1}) = ({}^t\sigma)^{-1}$, $(\bar{\sigma})^{-1} = (\sigma^{-1})^{-1}$

Remark 0.3. If σ, τ are regular matrices then $\sigma\tau$ are regular matrices.

* $\det : M_n(\mathbb{C}) \rightarrow \mathbb{C}$ can be expressed as

$$\begin{array}{ccc} M_n(\mathbb{C}) & \xrightarrow{\det} & \mathbb{C} \\ \downarrow f & & \\ \mathbb{C}^{n^2} & \xrightarrow{p} & \mathbb{C} \end{array}$$

p is some polynomial function of n^2 variables, which is continuous. We know f is continuous. Hence, $\det$ map is continuous.

Remark 0.4. $GL_n(\mathbb{C}) = \{\sigma \in M_n(\mathbb{C}) \mid \det \sigma \neq 0\}$

$$= M_n(\mathbb{C}) \setminus \det^{-1}\{0\}$$

is a group and an open subset as $\det$ is a continuous map.

(ii) For $\sigma = (a_{ij}) \in GL_n(\mathbb{C})$ the coefficient b_{ij} of σ^{-1} are given by $b_{ij} = \frac{1}{\det \sigma} A_{ij}$ where A_{ij} is a polynomial of σ . Hence mapping $\sigma \mapsto \sigma^{-1}$ is continuous on $GL(n, \mathbb{C})$. Moreover, it's a homeomorphism.

Remark 0.5. If $\sigma \in GL_n(\mathbb{C})$, we define $\sigma^* = ({}^t\sigma)^{-1}$. Then we have $(\sigma\tau)^* = \sigma^*\tau^*$. Hence $\sigma \mapsto \sigma^*$ is a homeomorphism, and an automorphism of order 2 of $GL_n(\mathbb{C})$.

Definition 0.1. $O_n(\mathbb{C}) = \{\sigma \in GL_n(\mathbb{C}) \mid \sigma^* = \sigma^{-1}\}$

is set of orthogonal matrices. If $\sigma = \sigma^*$ then σ is complex orthogonal and if $\sigma = \sigma^* = \bar{\sigma}$ then real orthogonal.

Definition 0.2. $U_n(\mathbb{C}) = \{\sigma \in GL_n(\mathbb{C}) \mid \sigma^* = \sigma^{-1}\}$

is set of all unitary matrices.

Note $U_n(\mathbb{C})$ or $O_n(\mathbb{C})$ are non empty as $e \in U_n(\mathbb{C})$ and $e \in O_n(\mathbb{C})$.

Notation: $U_n(\mathbb{C}) = (UC_n)$ and $O_n(\mathbb{R}) = OC_n$. Don't be confused!

Lemma 0.4. If f, g are continuous mapping from topological space X to Hausdorff space Y then

$$\{x \in X \mid f(x) = g(x)\}$$

is closed.

Proof. Consider $\{x \mid f(x) \neq g(x)\}$. Now

$$x \in \{x \in X \mid f(x) \neq g(x)\}$$

implies $f(x) \neq g(x)$ in Y . By Hausdorff property there exists open set U_1 and U_2 such that $f(x) \in U_1$ and $g(x) \in U_2$ and $U_1 \cap U_2 = \emptyset$. Now $f^{-1}(U_1)$ and $g^{-1}(U_2)$ are open. Let $V = f^{-1}(U_1) \cap g^{-1}(U_2)$ be a neighborhood of x . For any $y \in V$ $f(y) \neq g(y)$ otherwise $U_1 \cap U_2 \neq \emptyset$. Hence $x \in V \subseteq \{x \mid f(x) \neq g(x)\}$ is open. Hence $\{x \mid f(x) = g(x)\}$ is closed. $\square$

Proposition 0.1. $O_n(\mathbb{C})$, $U(n)$, $O(n)$ are closed subgroups of $GL_n(\mathbb{C})$.

Proof. Observe mapping $\sigma \mapsto \sigma^*$, $\sigma \mapsto \bar{\sigma}$ and identity mapping are continuous. By the above lemma, as $\mathbb{R}$ is T_2 space. The sets $U(n) = \{\sigma \mid \bar{\sigma} = \sigma^*\}$ and $O_n(\mathbb{C}) = \{\sigma \mid \sigma = \sigma^*\}$ are closed sets. Observe $GL_n(\mathbb{R}) = \{\sigma \mid \sigma = \bar{\sigma}\}$ is also closed. Hence $U(n)$ and

$$O(n) = O_n(\mathbb{C}) \cap GL_n(\mathbb{R})$$

is closed. As all the given sets are non-empty by subgroup test, we can show these are subgroups. $\square$

$$* \quad SL_n(\mathbb{C}) = \{\sigma \in GL_n(\mathbb{C}) \mid \det \sigma = 1\}$$

is special linear group.

As $\varepsilon \in SL_n(\mathbb{C})$, and by subgroup tests we can show $SL_n(\mathbb{C})$ is a subgroup of $GL_n(\mathbb{C})$. Moreover, $SL_n(\mathbb{C}) = \det^{-1}\{1\}$ implies $SL_n(\mathbb{C})$ is a closed subgroup.

Remark 0.6.

$$SL(n) = GL_n(\mathbb{C} \cap \mathbb{R}) \cap SL_n(\mathbb{C})$$

$$SU(n) = SL_n(\mathbb{C}) \cap U(n)$$

$$SO(n) = SL_n(\mathbb{C}) \cap O(n)$$

are closed subgroups of $GL_n(\mathbb{C})$.

Theorem 0.1. $U(n), O(n), SU(n)$ and $SO(n)$ are compact.

Proof. Observe $O(n), SU(n)$ and $SO(n)$ are closed subgroups of $U(n)$. It is suffice to show $U(n)$ is compact.

Goal: $U(n)$ is homeomorphic to a closed and bounded set of $\mathbb{C}^{n^2}$.

Observe $GL_n(\mathbb{C})$ is a open subgroup of $M_n(\mathbb{C})$ hence it is also a closed subgroup according to a previous lemma. Observe $U(n)$ is closed in $GL_n(\mathbb{C})$ and $GL_n(\mathbb{C})$ is closed in $M_n(\mathbb{C})$ implies $U(n)$ is closed in $GL_n(\mathbb{C})$.

Now for $\sigma = (a_{ij}) \in U(n)$ iff and only if

$${}^t \overline{\sigma} \sigma = \epsilon \iff \sum_k a_{ki} \overline{a_{kj}} = \delta_{ij}$$

implies $\sum_{k=1}^n |a_{ki}|^2 = 1$ which means for i^{th} column vector, the absolute sum is 1. This implies for any $1 \leq i, j \leq n$ $|a_{ij}| \leq 1$.

Now in $(C^{n^2}, \|\cdot\|_1)$, $v = (v_k)_{k=1}^{n^2}$, $|v_k| \leq 1$ for $1 \leq k \leq n^2$ we define $\|v\|_1 = \max_{1 \leq k \leq n^2} |v_k|$. Observe $f : M_n(\mathbb{C}) \rightarrow C^{n^2}$ as homeomorphism. $f(U(n))$ is closed and bounded, and compact. This implies $U(n)$ is compact.

□

Note: The choice of the norm does not matter as all norms in finite dimension vector space are equivalent (Add the proof to that in the appendix).